

Phase Noise

Origin

by

$$v_0(t) = V_0 [1 + A(t)] \cos[\omega_0 t + \theta(t)]$$

Where $A(t)$ represents the amplitude fluctuations of the output, and $\theta(t)$ represents the phase variation of the output waveform.

- Of these, **amplitude variations** can usually be well-controlled, and generally have less impact on system performance.
 - An instantaneous **phase variation** is indistinguishable from a variation in frequency.
- Small changes in the oscillator frequency can be represented as a frequency modulation of the carrier by letting

$$\theta(t) = \frac{\Delta f}{f_m} \sin \omega_m t = \theta_p \sin \omega_m t$$

where $f_m = \omega_m / 2\pi$ is the modulating frequency. The peak phase deviation is $\theta_p = \Delta f / f_m$.

- Set $A(t)=0$ to ignore amplitude fluctuations,

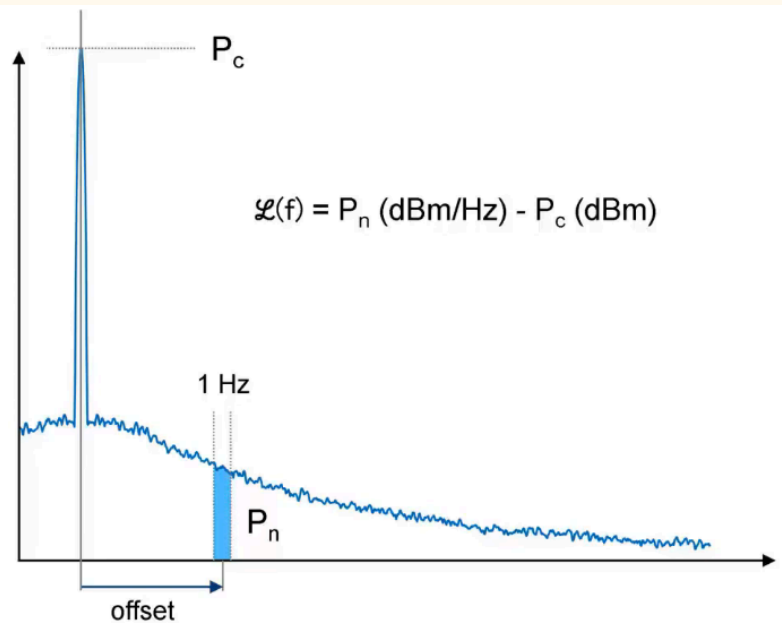
$$v_0(t) = V_0 [\cos \omega_0 t \cos(\theta_p \sin \omega_m t) - \sin \omega_0 t \sin(\theta_p \sin \omega_m t)]$$

Assuming the phase deviations are small, so that $\theta_p \ll 1$. $\sin(x) \approx x$, and $\cos(x) \approx 1$.

$$\begin{aligned} v_0(t) &= V_0 [\cos \omega_0 t - \theta_p \sin \omega_0 t \sin \omega_m t] \\ &= V_0 \left\{ \cos \omega_0 t - \frac{\theta_p}{2} [\cos(\omega_0 - \omega_m)t - \cos(\omega_0 + \omega_m)t] \right\} \end{aligned}$$

Measurement

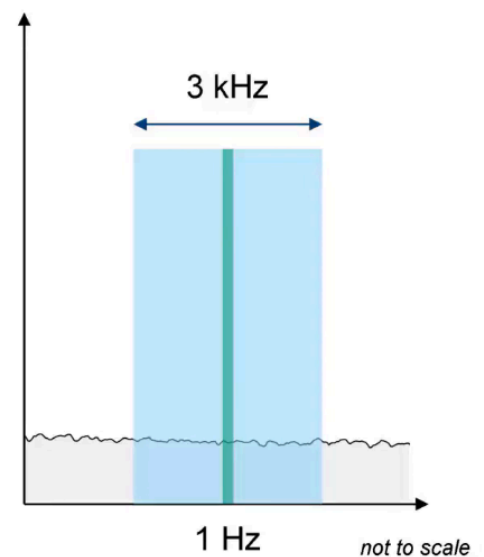
- ▶ Measure carrier power (P_c in dBm)
- ▶ Move to given offset
- ▶ Measure phase noise power (P_n)
 - Over 1 Hz bandwidth (dBm / 1 Hz)
- ▶ Subtract the power levels ($P_n - P_c$)
- ▶ Result is phase noise (dBc / Hz) at given offset
- ▶ Display results (plot and/or spot noise)
- ▶ Note that we must also:
 - Normalize
 - Correct shape



- ▶ Phase noise is specified as the power within a bandwidth of 1 Hz (dBc / 1 Hz)
- ▶ Power measured using a resolution bandwidth (RBW) filter
 - Usually more than 1 Hz wide
- ▶ Measured noise power is normalized to a 1 Hz bandwidth by subtracting N dB, where:

$$N = 10 \log RBW(Hz)$$

- ▶ Example:
 - Using a 3 kHz RBW, measured noise power is -90 dBm
 - $N = 10 * \log(3000) = 34.77 \text{ dB}$
 - Normalized 1 Hz noise power is -124.77 dBm



- ▶ Resolution BW filters are not perfectly rectangular
 - Usually have a Gaussian shape
- ▶ In addition to **normalizing** for **bandwidth**, we must also **correct** for the **shape** of the filter
- ▶ Gaussian has a wider noise BW than the nominal 3 dB BW
- ▶ Correction factor is required
 - Different depending on analyzer implementation
- ▶ Example:
 - Shape correction for this particular 3 kHz filter is 1.165
 - $N = 10 * \log(3000 * 1.165) = 35.43 \text{ dB}$
- ▶ Can be automatically done using a “noise marker”

